

Module 3: Singular Value Decomposition

Remark. The goal of these module is to prove the Singular Value Decomposition (SVD), which is the analogue of the Spectral Theorem we saw last time for arbitrary matrices (i.e., they can be rectangular, and no assumption is made on their rank or other structure such as symmetry). Here is the full statement of the theorem we want to prove.

1 Singular Value Decomposition.

Let $A \in \mathbb{R}^{m \times n}$. There exist orthogonal matrices $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ and a matrix $\Sigma \in \mathbb{R}^{m \times n}$ with nonnegative entries only along its main diagonal such that

$$A = U \Sigma V^T = \sum_{i=1}^{\rho} \sigma_i U_{:,i} V_{:,i}^T$$

where $\rho = \min\{m, n\}$. Moreover, we may take $\rho = \text{rank}(A)$.
To begin, we need one more ingredient about eigenvalues of symmetric matrices.

Definition. A symmetric matrix $A \in \mathbb{R}^{n \times n}$ is positive definite provided $\langle Ax, x \rangle > 0$ for all $x \in \mathbb{R}^n$. If instead $\langle Ax, x \rangle \geq 0$ for all $x \in \mathbb{R}^n$, the matrix is called positive semidefinite. For short, we will call symmetric, positive definite (respectively, symmetric positive semidefinite) matrices SPD (respectively, SPSD) for short.

1.1 Proposition

Any matrix of the form AA^T or $A^T A$ is SPSD.

1.1.1 Proof:

Simply note that $\langle A^T A x, x \rangle = \langle Ax, Ax \rangle = |Ax|^2 \geq 0$ (this could be 0 if A has nontrivial nullspace). For AA^T , simply note this is of the form $B^T B$ where $B = A^T$ and the same argument works.

1.2 Proposition

If $A \in \mathbb{R}^{n \times n}$ is SPSD, then its eigenvalues are real and nonnegative.

1.2.1 Proof

That the eigenvalues are real follows from Proposition 4.6. To see positivity, suppose that (λ, v) is an eigenvalue/eigenvector pair of A . Then $\lambda \langle v, v \rangle = \langle \lambda v, v \rangle = \langle Av, v \rangle$, whence because $v \neq 0$,

$$\lambda = \frac{\langle Av, v \rangle}{\langle v, v \rangle} \geq 0$$

where the inequality follows from the fact that A is positive semi-definite.

1.2.2 Proof of the SVD

To begin, consider the $A^\top A \in \mathbb{R}^{n \times n}$, which is SPSD by Proposition 5.3, and hence has real, nonnegative eigenvalues and we say $\lambda_1, \dots, \lambda_k$, with corresponding orthonormal eigenvectors $v_1, \dots, v_k$.

Now we note that $A^\top A$ and $AA^\top$ have the same eigenvalues. More precisely, the following holds.

1.3 Lemma

Let $A \in \mathbb{R}^{m \times n}$ be arbitrary. If (λ, v) is an eigenvalue/eigenvector pair of $A^\top A$, then (λ, Av) is an eigenvalue/eigenvector pair of $AA^\top$.

1.3.1 Proof:

By assumption, $A^\top Av = \lambda v$, whereupon multiplication by A yields $(AA^\top)Av = \lambda(Av)$. Since 0 cannot be an eigenvector, we must confirm that v is not in the null space of A , but this follows from the fact that $N(A) = N(A^\top A)$. Now, since we have the eigenpairs of $A^\top A$ in hand, it follows that,

$$u_i := \frac{Av_i}{\sqrt{\lambda_i}}, \quad i = 1, \dots, k \quad (3)$$

since $u_1, \dots, u_k$ are still eigenvectors of $AA^\top$ with associated eigenvalues $\lambda_1, \dots, \lambda_k$. Now we claim that $\{u_i\}_{i=1}^k$ is an orthonormal set.

1.4 Lemma

Let $\{u_i\}_{i=1}^k$ be defined as in (3). Then $\{u_i\}_{i=1}^k$ is an orthonormal set.

1.4.1 Proof:

Let i and j be fixed. By definition,

$$\langle u_i, u_j \rangle = \left\langle \frac{1}{\sqrt{\lambda_i}} Av_i, \frac{1}{\sqrt{\lambda_j}} Av_j \right\rangle = \frac{1}{\sqrt{\lambda_i \lambda_j}} \langle Av_i, Av_j \rangle = \frac{1}{\sqrt{\lambda_i \lambda_j}} \langle v_i, A^\top Av_j \rangle$$

Since (λ_j, v_j) is an eigenvalue/eigenvector pair for $A^\top A$,

$$\langle u_i, u_j \rangle = \frac{1}{\sqrt{\lambda_i \lambda_j}} \langle v_i, \lambda_j v_j \rangle = \frac{\sqrt{\lambda_j}}{\sqrt{\lambda_i}} \langle v_i, v_j \rangle = \begin{cases} 0, & i \neq j \\ 1, & i = j \end{cases}$$

The final calculation follows from the fact that $\{v_i\}_{i=1}^k$ is orthonormal, and the fact that $\{u_i\}_{i=1}^k$ is orthonormal follows directly from the above calculation. Now letting $\sigma_i = \sqrt{\lambda_i}$ for $i = 1, \dots, k$, define

$$U_k := \begin{bmatrix} | & & | \\ u_1 & \cdots & u_k \\ | & & | \end{bmatrix}, \quad v_k := \begin{bmatrix} | & & | \\ v_1 & \cdots & v_k \\ | & & | \end{bmatrix}, \quad \Sigma_k := \begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_k \end{bmatrix}$$

1.5 Lemma

With the above definitions, we have $U_k^\top AV_k = \Sigma_k$.

1.5.1 Proof

Note that by the inner product definition of matrix multiplication,

$$U_k^\top AV_k = \begin{bmatrix} \langle Av_1, u_1 \rangle & \langle Av_2, u_1 \rangle & \cdots & \langle Av_k, u_1 \rangle \\ \langle Av_1, u_2 \rangle & \langle Av_2, u_2 \rangle & \cdots & \langle Av_k, u_2 \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle Av_1, u_k \rangle & \langle Av_2, u_k \rangle & \cdots & \langle Av_k, u_k \rangle \end{bmatrix}$$

Now by definition of u_i and σ_i , we have

$$\langle Av_i, u_j \rangle = \left\langle Av_i, \frac{Av_j}{\sigma_j} \right\rangle = \frac{1}{\sigma_j} \langle Av_i, Av_j \rangle = \frac{1}{\sigma_j} \langle v_i, A^\top Av_j \rangle = \frac{\lambda_i^2}{\sigma_j} \langle v_i, v_j \rangle$$

Thus by orthonormality of $\{v_i\}$, we have

$$\langle Av_i, u_j \rangle = \begin{cases} 0, & i \neq j \\ \sigma_i, & i = j \end{cases} \quad (4)$$

The desired conclusion follows from (4).

Remark. *Proof of the SVD. Our proof of the SVD is almost complete. First, we complete $\{u_1, \dots, u_k\}$ to an orthonormal basis of $\mathbb{R}^{m \times m}$, denoted $\{u_1, \dots, u_m\}$, and likewise complete $\{v_1, \dots, v_k\}$ to an orthonormal basis of $\mathbb{R}^{n \times n}$, say $\{v_1, \dots, v_n\}$. We also make Σ an $m \times n$ matrix by adding zeros to make up the rest of the entries.*

Our final claim is that $U^\top AV = \Sigma$. To see this, first note that $v_{k+1}, \dots, v_n \in N(A)$. This implies that if $i > k$, then the i, j -th entry of $U^\top AV$ is $\langle Av_i, u_j \rangle = 0$. For $1 \leq i \leq k$ and $1 \leq j \leq k$, we have already seen that $\langle Av_i, u_j \rangle$ is either 0 or σ_i by (4). We only need to check that for $1 \leq i \leq k$ and $j > k$ that $\langle Av_i, u_j \rangle = 0$

to verify the claim. In this case, we go back to the definition of u_j and use the same argument as in to see that

$$\langle Av_i, u_j \rangle = \langle \sigma_i u_i, u_j \rangle = 0$$

The conclusion follows that $U^\top AV = \Sigma$.

The following calculation, which follows by orthogonality of the matrices U and V , finishes the proof:

$$U\Sigma V^\top = UU^\top AVV^\top = A$$

we find that if $A = U\Sigma V^\top$, then

$$A = \sum_{i=1}^{\text{rank}(A)} \sigma_i u_i v_i^\top \quad (5)$$

As a matter of terminology, we call the columns of U the left singular vectors of A , the columns of V the right singular vectors of A , and $\sigma_1, \dots, \sigma_{\text{rank}(A)}$ the singular values of A . Additionally, we assume the convention that the singular values are in descending order so that $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_{\text{rank}(A)} \geq 0$.

Now let's take a step back and consider what the SVD is really telling us. Let us consider the geometry of a linear transformation (matrix) $A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as shown in Figure 2.

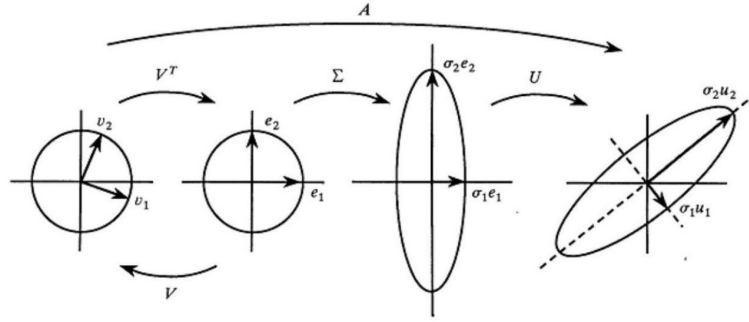


Figure 2.

Geometric representation of the SVD.

Figure from <https://blogs.sas.com/content/iml/2017/08/28/singular-value-decomposition-svd-sas.html>

Figure 2 illustrates the fact that $V^\top$ maps the generic orthonormal basis $\{v_1, v_2\}$ onto the canonical basis $\{e_1, e_2\}$. Then Σ stretches e_1 and e_2 , and U is another rotation onto a new orthonormal basis. Hence the SVD for square matrices can be seen as the sequence: rotate, stretch, rotate again. In higher dimensions, rotation means multiplication by orthogonal matrices with determinant 1.

Another thing to note about what the SVD is doing algebraically is that it finds two orthogonal bases in which the transformation A mapping the basis $\{v_1, v_2\}$ to $\{u_1, u_2\}$ is diagonal. In particular, we have that $Av_i = \sigma_i u_i$.

2 Python Programming Tasks

2.1 Task 1: Implement Singular Value Decomposition in Python

Goal: Implement SVD in Python using NumPy's built-in functions and analyze the results.

2.1.1 Instructions:

1. Implement a Python function that performs SVD on a matrix using `numpy.linalg.svd()`. This function should:
 - Take a matrix A as input.
 - Return the matrices U , Σ , and V^T .
2. Test your implementation on the following matrix:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

3. Print out the resulting matrices U , Σ , and V^T , and verify that the product UV^T reconstructs the original matrix A .

Expected Outcome: A Python implementation of SVD that returns the decomposed matrices and verifies that the reconstruction of A is accurate.

2.2 Task 2: Visualize the Singular Values and Their Contribution

Goal: Visualize the singular values and understand their significance in matrix approximation.

2.2.1 Instructions:

1. Write Python code to:
 - Perform SVD on a matrix A using `numpy.linalg.svd()`.
 - Plot the singular values from the diagonal matrix Σ using `matplotlib`.
2. Analyze the contribution of each singular value to the overall matrix by:

- Truncating smaller singular values.
 - Reconstructing the matrix using only the top k singular values, where k is a smaller number than the rank of A .
3. Test this on the matrix:

$$A = \begin{bmatrix} 3 & 2 & 2 \\ 2 & 3 & -2 \end{bmatrix}$$

and plot the matrix approximations for different values of k .

Expected Outcome: A plot showing the singular values and matrix approximations for different values of k , illustrating the role of singular values in capturing the essential structure of a matrix.

2.3 Task 3: Apply SVD to Dimensionality Reduction (Principal Component Analysis)

Goal: Use SVD for dimensionality reduction, specifically Principal Component Analysis (PCA).

2.3.1 Instructions:

1. Download a standard dataset such as the Iris dataset.
2. Perform SVD on the centered data matrix (after subtracting the mean from each feature):
 - Decompose the matrix into U , Σ , and V^T using SVD.
 - Use the top k singular values and corresponding singular vectors to project the data into k -dimensional space.
3. Visualize the 2D projection of the Iris dataset using the top 2 principal components (i.e., the top 2 singular values).

Expected Outcome: A 2D scatter plot showing the Iris dataset projected onto its top two principal components, along with an explanation of how SVD is applied in PCA.

2.4 Task 4: Image Compression Using Singular Value Decomposition

Goal: Apply SVD to image compression by reducing the rank of the matrix representation of an image.

2.4.1 Instructions:

1. Load a grayscale image into Python using libraries such as matplotlib, image or PIL.
2. Treat the image as a matrix A of pixel intensities and perform SVD:

$$A = U\Sigma V^T$$

3. Reconstruct the image using only the top k singular values, where k is less than the full rank of the matrix. Test for different values of k , such as $k = 5, 20, 50$, etc.
4. Plot the original image and the compressed versions using matplotlib.

Expected Outcome: Original and compressed versions of the image displayed side by side, with a discussion on the trade-off between image quality and the number of singular values used.

2.5 Task 5: Solve a Linear System Using SVD

Goal: Use SVD to solve a system of linear equations.

2.5.1 Instructions:

1. Consider the linear system $Ax = b$, where A is a 3×3 matrix and b is a vector:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 10 \end{bmatrix}, \quad b = \begin{bmatrix} 3 \\ 3 \\ 4 \end{bmatrix}$$

2. Solve for x using SVD. First, decompose A into U, Σ , and V^T , and then use the decomposition to find the solution.
3. Compare your solution to that obtained using `numpy.linalg.solve()`.

Expected Outcome: A solution to the linear system using SVD and verification by comparing the results with other standard methods.

2.6 Task 6: Explore SVD for Low-Rank Approximation

Goal: Understand how SVD can be used for low-rank approximation of a matrix and its application in noise reduction.

2.6.1 Instructions:

1. Generate a random matrix A of size 5×5 and add some random noise to it.
2. Perform SVD on the noisy matrix and use only the top singular values to reconstruct a low-rank approximation of A .
3. Compare the original matrix, noisy matrix, and the denoised matrix using visualizations and matrix norm calculations (such as the Frobenius norm).

Expected Outcome: A comparison between the noisy matrix and the low-rank approximation, illustrating how SVD can be used for noise reduction.